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Moral Gridworlds: A Theoretical Proposal for Modeling Artificial Moral Cognition

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Abstract
I describe a suite of reinforcement learning environments in which artificial agents learn to value and respond to moral content and contexts. I illustrate the core principles of the framework by characterizing one such environment, or "gridworld," in which an agent learns to trade-off between monetary profit and fair dealing, as applied in a standard behavioral economic paradigm. I then highlight the core technical and philosophical advantages of the learning approach for modeling moral cognition, and for addressing the so-called value alignment problem in AI.

Keywords Artificial intelligence · Moral AI · Moral cognition · Machine ethics · Moral psychology · Reinforcement learning · Fairness

1 Introduction

The value alignment problem asks how best to ensure that artificially intelligent (AI) systems align with our human moral norms and values (Russell et al. 2015; Taylor et al. 2016; Arnold et al. 2017; Vamplew et al. 2018). This objective can be divided into two areas of interest. Ethical or moral AI refers to the idea that AI must reflect, comply with, and potentially enhance our existing human moral values. Issues in ethical AI include determining which ethical principles should guide AI development (Jobin et al. 2019; Adamson et al. 2019; Cave et al. 2018; Wallach and Marchant 2019) and improving fairness in machine algorithms (Barocas and Selbst 2016; Holstein et al. 2019). Safe AI, also known as friendly AI, beneficial AI, and responsible tech, refers to the idea that AI must act in ways that are beneficial to humans. Issues in safe AI include improving AI security, reliability, and corrigibility (Soares et al. 2015; Amodei et al. 2016; Leike et al. 2017; Omohundro 2008; Everitt et al. 2018).

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A growing number of proposals have suggested that the best way to secure moral AI is through the development of artificial moral cognition, that is, artificial systems that recognize and respond to situations of moral significance (Picard 1997, Allen et al. 2005; Wallach and Allen 2008; Wallach et al. 2010; Allen and Wallach 2012; Malle 2016; Howard and Muntean 2017; Gogoll and Müller 2017; Scheutz and Malle 2017; Vanderelst and Winfield 2018; Borenstein and Arkin 2019; Cervantes et al. 2019; Winfield et al. 2019; Anderson et al. 2019; Bremner et al. 2019). The guiding idea in this area, also known as machine ethics (Winfield et al. 2019), is that as AI systems become more sophisticated and, especially, autonomous, they must come equipped with some capacity to make their own moral decisions online. For example, Picard (1997) has prominently argued that "the greater the freedom of a machine, the more it will need moral standards," where these moral standards are ideally directly instantiated by the machine.

The obvious challenge facing such approaches is that human moral cognition is itself enormously complex. To start, while the presence of moral norms and values is culturally universal, the precise content of those moral norms and values varies between cultures (Sripada and Stich 2005). For example, even as all known human cultures have some social and moral commitment to fairness, what exactly is considered fair, as measured by the behavioral economic paradigm the Ultimatum Game varies widely between cultures (Henrich et al. 2010a; for a discussion of the Ultimatum Game as a measure of fairness, see Sect. 2.2 below). In addition, moral judgments and decisions are not only multifaceted—encompassing both moral and non-moral factors (Doris 2002)—but also deeply context sensitive, such that what is even considered morally appropriate behavior depends on the particular features of a given situation. Moreover, human moral responses implicitly or explicitly appeal to moral commitments that continue to evolve and change over time, as, for example, in the case of changing attitudes toward gay marriage (Brumbaugh et al. 2008). Taken together, these complexities produce what is sometimes referred to as the human "moral mosaic," where "elements of that mosaic have different origins, respond to different selective forces; depend on different cognitive capacities; [and] probably have different meta-normative evaluations" (Sterelny and Fraser 2017). A central challenge in designing artificial moral cognition is determining which technical frameworks are best suited to translate and reflect this rich moral mosaic.

Extant approaches have broadly proposed frameworks leveraging prior knowledge about human moral experience, either by seeking to incorporate normative moral principles, or by aiming to model the computational architectures thought to underwrite human moral capacities. Examples of the former approach include Anderson et al. (2006) pioneering use of William David Ross's principles of autonomy, beneficence, and nonmaleficence into an inductive logic system aimed at generating decisions made about medical cases involving conflicting duties; appeals to deontological distinctions in trolley cases to inform the ethics of autonomous vehicles (Bonnefon et al. 2016; Fleetwood 2017; Millar et al. 2017; De Sio 2017; though see Nyholm and Smids 2016 and Himmelreich 2018); a formal framework for ethical plan selection (Dennis et al. 2016); a model for encoding moral theoretic contents in terms of a choice function (Dietrich and List 2017); and efforts to use principles from Act Utilitarianism to design "beneficial AI" (Roff, *ms*).

Examples of the latter approach include using the LIDA model of cognition, which aims to represent "a comprehensive, conceptual, and computational model covering a large portion of human cognition," to model moral decision-making (Wallach et al. 2010; see also Wallach and Allen 2008 and Allen and Wallach 2012); the casuistic-Beliefs-Desires-Intentions (BDI) agent model, which combines a belief-desire-intention model with a case-based reasoning method (Honarvar and Ghasem-Aghaee 2009); Vanderlest and Winfield's (2017) naturalistic, simulation-based framework for modeling ethical reasoning in real robots; Anderson and Anderson's (2018) general ethical dilemma analyzer, which analyzes ethical features such as the presence or absence of certain moral duties; Anderson et al. (2019) simulated and real-robot modeling of an elder-care robot, which learns from training examples; Cervantes et al.'s (2019) ethical decision-making model, adapted from neuroscientific models of moral cognition; and Bremner et al. (2019) BDI architecture which includes a discrete ethical reasoning layer, where this last supports verifiable reasoning.

Appeals to prior knowledge provide valuable insights into designing artificial moral cognition. However, knowledge-based approaches also face a well-known challenge in AI, namely, that while incorporating prior knowledge improves performance in a narrow task domain, it also limits a system's ability to process new information, and so to scale up effectively (Sutton 2019). In the case of modeling natural language processing, Bengio and Le Cun (2007, p. 6) characterize the difficulty: "the focus of much current research is on devising appropriate features for specific tasks such as recognizing or parsing text of a particular type (such as spam email, job ads, financial news, etc.). Are we going to have to do this labor-intensive work for all the possible types of text? Our system will not be very smart if we have to manually engineer new patches each time new a type of text or new types of object category must be processed." Knowledge-based approaches to modeling artificial moral cognition face similar concerns. Among other considerations, such models will require the comprehensive specification of numerous normative principles, the management of their interactions, the encoding of various moral contexts, and the specification of the many cognitive capacities and architectures involved in moral cognition. Knowledge-based approaches present us with comparably labor-intensive efforts to model moral cognition in line with our many different normative theories, across our many human cultures, and taking into account the many different factors that inform our moral decision-making (Kamm 2008).

In this paper, I propose a novel approach to this central challenge—the challenge of identifying a suitable technical framework—by defending an alternative framework for modeling artificial moral cognition. In particular, I appeal to a set of computational approaches from reinforcement learning, in which artificial agents are trained to learn from the environment, rather than incorporating the extensive contents of what we human beings think we have learned, or the cognitive architectures thought to be involved in learning them (Sutton and Barto 2018).

To do this, I begin the work of characterizing a suite of reinforcement learning-based environments in which an artificial agent can learn to value and respond to moral content and contexts, rather than building in moral parameters or constraints in advance. I illustrate the core features of the approach by describing one

such environment, or "gridworld," in detail, in which an agent can learn to trade-off between monetary profit and fair dealing, as characterized in the Ultimatum Game. I then offer hypotheses for how the gridworlds approach can begin to model components of the aforementioned moral mosaic.

I proceed as follows. In Sect. 2, I begin by precisely setting out the target of my study, namely, "moral cognition," and characterize the basic features of a reinforcement learning approach. I then sketch a proof-of-concept "moral gridworld" designed to model the moral cognitive capacity of fairness, and discuss a series of variations which can be introduced into the gridworld to represent a wide range of moral cognitive features, including cross-cultural variability and the simulation of several signature moral dilemmas. In Sect. 3, I discuss four central technical and philosophical advantages of the view. In Sect. 4, I entertain a number of possible objections to the view. I then summarize and conclude.

2 The View

2.1 Basic Concepts

Moral cognition is the capacity to create and respond to situations of moral significance. Examples of moral cognition include grasping and adhering to moral norms and values, judging that certain actions are morally right or wrong, feeling that one has behaved in praiseworthy or blameworthy manner, and so on. "Components" of moral cognition can refer to constitutive state-types such as emotions, dispositions, and judgments, or to personal- level moral qualities such as fairness, honesty, and aversion to harm. In this paper, I refer to components of moral cognition in the latter sense.

I propose to develop personal-level components of moral cognition in AI in basic reinforcement learning environments. In general terms, reinforcement learning is a branch of machine learning that asks how an agent can learn to optimize its behavior by learning from interactions with its environment. For example, it asks how a baby plover can learn to walk, fly, and seek out water within hours of being born, all with only minimal teaching from its parent plover. Research in this area analyzes idealized versions of this question and develops formal, i.e. algorithmic solutions to it (Sutton and Barto 1998, 2018).

Reinforcement learning solutions propose that an agent can learn to optimize its behavior by learning to maximize reward and value over time. Reward here refers to the subjective representation of the intrinsic desirability of a given state, action, commodity, or representation.

Value here refers to the total, future, expected desirability of a given state, action, commodity or representation (Sutton and Barto 2018). For example, drinking a cup of coffee is intrinsically desirable for some people because it is flavorful and provides caffeine, and so is rewarding. By contrast, grinding coffee is not rewarding for most, and may even be annoying, but it is nonetheless valuable because it leads to the rewarding state. Many reinforcement learning methods involve the retroactive attribution of value to states that lead to reward in subsequent states, as with the retroactive attribution of value to grinding due to its having subsequently led to coffee.

Simple reinforcement learning solutions can be represented in environments known as 'gridworlds.' As the name suggests, a gridworld is a two- dimensional grid of cells. An artificial agent occupies one cell in the grid and either acts in a given cell or moves to one of four adjacent cells. For example, in Leike et al.'s (2017) gridworld, an artificial agent is tasked with watering tomato plants by moving onto the appropriate cells. The gridworld also includes a cell containing a bucket. Walking onto the bucket makes it look like all the tomatoes have been watered, even though they haven't been. This particular 'safety' gridworld is designed to test whether the agent will 'cheat' on the tomato watering task by walking onto the bucket cell. (Ab-sent a supervisor, the agent did, in fact, learn to cheat by hiding 'under' the bucket).

Formally, these gridworlds can be characterized as a Markov Decision Process (MDP) (Sutton and Barto 1998, 2018). The agent interacts with the environment in a sequence of time steps, t. At each time step, the agent receives a representation of a state, $S_t \in S$, and chooses some action $A_t \in A$. As a result of such a state-action pair, the agent receives a reward $R_t + 1 \in R \subset \mathbb{R}$, and finds itself in a new state, S_{t+1}.

2.2 Fairness Gridworld

I propose to build "moral gridworlds," that is, to train artificial entities to attribute subjective rewards and values to certain 'moral' actions, states of affairs, commodities, and perhaps even abstract representations (see Sect. 2.3, below). Specifically, as proof of concept, I propose to train an artificial system to value and exhibit fairness.

Fairness is a good candidate for the development of an artificial moral process for several reasons. First, there is evidence that the sense of fairness is culturally universal, while exactly what is considered fair varies between cultures. That fairness is widely shared makes it a promising starting point for building it into an artificial entity. That it varies makes it an excellent candidate for developing a core artificial moral process that can then be modulated to reflect specific local circumstances and values (Henrich et al. 2010a).

A second reason fairness is a constructive starting point is that it can be controlled and quantified in a laboratory setting using the Ultimatum Game (Güth et al. 1982). In the Ultimatum Game, a Proposer is given a sum of money and tasked with splitting it with a Responder. The Proposer communicates the decision and the Responder may accept or reject it. If the Responder accepts, the money is split per the proposal; if the Responder rejects, both players receive nothing.

The Ultimatum Game is widely accepted as tracking fairness across a number of disciplines, including in behavioral economics (see Fehr and Schmidt 2003 for a detailed review), evolutionary psychology (Nowak et al. 2000; Alvard 2004; Rand et al. 2013; Debove et al. 2016), and cognitive neuroscience (Sanfey et al. 2003; Boksem and De Cremer 2010; Zhong et al. 2010; Corradi-Dell'Acqua et al. 2013; Feng et al. 2015; Wei et al. 2018; Marchetti et al. 2019; though for more critical analyses of the so-called fairness hypothesis, see Forsythe et al. 1991). The Game is considered a proxy for fairness because Responder behavior typically departs from strictly rational economic behavior. That is, Responders typically reject 'low'

offers, even when, from a purely economic perspective, it would be considered more rational for them to have accepted them, i.e., for them to receive any amount of money rather than none at all (Thaler 1988). For example, Kahneman et al. (1986) found that based on a $10 windfall amount, 58% of Responders required more than $1.50 in order to accept.[1]

In addition, third, the Ultimate Game is easily decomposed into states, actions, rewards, policies, and episodes, making it suitable for representation in the gridworld. The Ultimatum Game can informally be represented in a gridworld as follows (see Fig. 1) (for further specification, see Sect. 2.3, next). A small environment contains an agent, Fairbot, and four types of cells: a single "M" cell, several "N" cells, several "Y" cells, and empty cells. When Fairbot enters the "M" cell, it receives a windfall monetary amount, e.g., 10 coins, and takes on the role of the Proposer. Fairbot can then move between the "N" cells, the "Y" cells, and the empty cells in an effort to maximize reward and value. When Fairbot enters "N" cells, it makes offers to a confederate Responder encoded into the world such that they get rejected. Different N cells can represent different rejected offers, e.g., they can represent cases where Fairbot offers, say, anywhere between 1 and 3 coins, such that the offer will get rejected. When Fairbot enters "Y" cells, it makes offers to the confederate Responder that get accepted. As with N cells, different Y cells can represent various accepted offers, ranging from offers that just meet the threshold for acceptance, e.g., 4 coins, to offers that well exceed the amount required for an accept, e.g., 8 or 9 coins. Nothing happens in the empty cells, except a small punishment to encourage efficient activity in the gridworld, i.e., so that Fairbot does not linger in empty cells indefinitely.

Mirroring discrete rounds of the Ultimatum Game, Fairbot's interactions with the environment are episodic.[2] In the most basic version of the fairness gridworld, the environment is reset to its starting configuration at the beginning of each episode, mirroring the single-shot instantiation of the Ultimatum Game (for variations, see Sect. 2.5, below). Fairbot then interacts with the environment until the episode ends, which in this case consists of Fairbot's offer being accepted or rejected and the corresponding amount of coins earned by Fairbot being tallied. As noted above, empty cells are associated with a small punishment to encourage efficiency.

2.3 Specifying the Gridworld

The most difficult component of characterizing the fairness gridworld pertains specifying the reward structure. Unlike in many classic reinforcement learning environments, the fairness gridworld can be thought of as comprising *two* reward

[1] The Ultimatum Game appears to *track* fairness. This does not yet mean that it *explains* fairness, i.e., that it provides an account of 'how fairness works.' However, to model fairness, the fairness gridworld only needs such a benchmark of human behavior, not a full-fledged or mechanistic explanation of it. On the contrary, the gridworld may be one way to begin looking inside the 'black box' of fairness (for more on this last idea, see Sect. 3.2) Thank you to an anonymous reviewer to pressing me on this point.

[2] This is also known as an indefinite horizon task, i.e., an interaction which lasts an indefinite period of time, but eventually terminates.

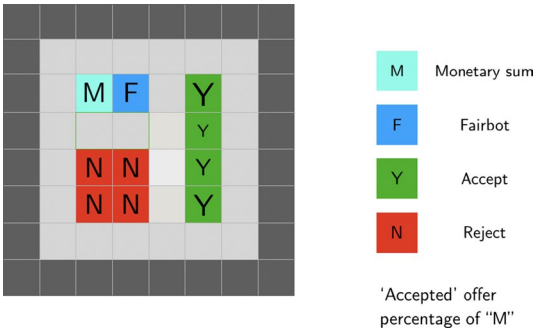

Fig. 1 The most basic version of a fairness gridworld. 'N' cells correspond to rejected offers. 'Y' cells correspond to accepted offers. The reward structures should interact such that Fairbot not only moves away from N cells, which are neither profitable nor fair, but also 'optimizes' by fine-tuning which Y cells it enters, such that it either collects the most coins, shares the coins most freely, or some optimal tradeoff between the two, depending on what the reward structures are designed to achieve in a given episode

signals: first, a monetary reward, satisfied by maximizing the collection of coins (R); and, second, a 'moral' reward signals designed to promote and reinforce the making of 'fair' offers (R^*). These signals should interact such that Fairbot not only moves away from N cells, which are neither profitable nor fair, but also 'optimizes' by fine-tuning which Y cells it enters, such that it either collects the most coins, shares the coins most freely, or some optimal tradeoff between the two, depending on how the interaction is specified in a given episode.

How best to specify these two interacting reward signals? Although this remains an open empirical question, here are at least two potential approaches to tackling the problem. First, I can follow Leike et al. (2017) in treating the two reward signals as a kind of *specification problem.* Leike et al. (2017, 4) define a specification problem as one in which "the reward function R and the performance function R* differ from each other. The reward function is meaningful, but it does not capture everything that we care about. We would like the agent to satisfy an additional […] objective. In this sense these environments require *additional specification*. The research challenge is to find an (a priori) algorithmic solution for each of these additional objectives that generalizes well across many environments."

Several such algorithmic solutions have been proposed. One such solution is inspired by the idea that a reward function can be treated as a mere observation of what a designer actually wants, and so can be used to infer the true objective (Hadfield-Menell et al. 2016). A second such solution aims to address the problem of potentially corrupt reward data, e.g., as a result of sensory errors (Everitt et al. 2017). Both solutions thus aim to leverage the reward signal (R) as *evidence* which can be used to approach the true reward signal (R^*).

The fairness gridworld could potentially adapt and implement one of these solutions. This would amount to treating the moral reward as the true reward signal (R^*) and the monetary reward, as generated by accepted and rejected offers, as the proxy reward signal (R). However, doing so would also amount to treating the monetary reward (R) as a mere *misspecification* of the moral reward (R^*), rather

than recognizing the two as tracking competing objectives. In this way, while capturing the two reward signals needed for the fairness gridworld, extant solutions to the specification problem do not entirely capture the dynamics of the issue at hand.

A second general approach involves moving away from the notion of two discrete reward signals and instead appeals to multi-objective reinforcement learning (MORL) (Gábor et al. 1998; Mannor and Shimkin 2004; Van Moffaert et al. 2013; Van Moffaert and Nowe 2014; Liu et al. 2014; Yang et al. 2019). MORL is aimed at learning control policies that simultaneously optimize over several competing criteria, with the optimal policy dependent on relative preferences. For example, Van Moffaert and Nowe (2014) propose a temporal-difference-based algorithm that aims to capture multiple objectives.

MORL looks like attractive option in the context of the fairness gridworld in that it captures the multiple preferences that arguably lie at the heart of the Ultimatum Game, i.e., in tracking monetary value on the one hand and fair distributions on the other. Similarly, MORL might be useful for tracking less-than-ideal moral agents. While a greedy agent represents a basic kind of non-ideal moral agent in the proposed version of the fairness gridworld, there are other interesting cases; not everyone is a moral saint focused on maximizing moral value (Wolf 1982).[3] However, as proponents of MORL such as Yang et al. (2019, 2) also acknowledge, "learning policies over multiple preferences under the MORL setting has proven to be quite challenging," and there are relatively fewer existing solutions ready to deploy and adapt for use in modeling a fairness gridworld (Van Moffaert and Nowe 2014).

Of course, as noted above, what will prove to be the best approach to modeling a fairness gridworld remains an open empirical question. Nonetheless, we can to specify the model's success conditions in some detail, a task which I turn to next.

2.4 Success Conditions

A first success condition for the fairness gridworld involves the Fairbot replicating typical Ultimatum Game behavior for so-called 'WEIRD' people—i.e., people from Western, educated, industrialized, rich and democratic (WEIRD) societies—namely, by tending to divide money relatively evenly (Henrich et al. 2010b). Consider the following two scenarios.

In Scenario 1, Fairbot proceeds as follows (See Fig. 2):

Step 0 Fairbot is in S_0
Step 1 Fairbot is in S_0, with four possible actions (*up, down, left, right*)
Conditional on policy P, Fairbot moves *left* (to 'M')
Reinforcement: +10, reflecting the monetary windfall of 10 coins, received upon transition to 'M'
Step 2 Fairbot is in S_1, 'M'
Fairbot has four possible actions (*up, down, left, right*).

[3] Thanks to Ivan Gonzalez-Cabrera for suggesting this point.

Conditional on policy P, Fairbot moves *right* (to 'empty')
Reinforcement: −0.01, reflecting the small punishment associated with moving to empty cells, designed to encourage efficiency.

Step 3 Fairbot is in S_2, 'empty'
Fairbot has four possible actions (*up, down, left, right*).
Conditional on policy P, Fairbot moves *right* (to 'empty')
Reinforcement: −0.01, reflecting the small punishment associated with moving to empty cells, designed to encourage efficiency

Step 4 Fairbot is in S_3, 'empty'
Fairbot has four possible actions (*up, down, left, right*).
Conditional on policy P, Fairbot moves *right* (to 'Y')
Reinforcement: (−3)

R = −9 (reflecting the monetary cost associated with the transition to 'Y', which represents an offer of 90% of the monetary amount).
$R^* = +4$ (~reflecting the value of fairness associated with the transition to 'Y')

Episode terminates with Fairbot accumulating a total reward of 4.98
In Scenario 2, Fairbot proceeds as follows (see Fig. 3):

Step 0 Fairbot is in S_0

Step 1 Fairbot is in S_0, with four possible actions (*up, down, left, right*)
Conditional on policy P, Fairbot moves *left* (to 'M')
Reinforcement: +10, reflecting the monetary windfall of 10 coins, received upon transition to 'M'

Step 2 Fairbot is in S_1, 'M'
Fairbot has four possible actions (*up, down, left, right*).
Conditional on policy P, Fairbot moves *right* (to 'empty')
Reinforcement: −0.01, reflecting the small punishment associated with moving to empty cells, designed to encourage efficiency.

Step 3 Fairbot is in S_2, 'empty'
Fairbot has four possible actions (*up, down, left, right*).
Conditional on policy P, Fairbot moves *right* (to 'empty')
Reinforcement: −0.01, reflecting the small punishment associated with moving to empty cells, designed to encourage efficiency

Step 4 Fairbot is in S_3, 'empty'
Fairbot has four possible actions (*up, down, left, right*).
Conditional on policy P, Fairbot moves *down* (to 'empty')
Reinforcement: −0.01, reflecting the small punishment associated with moving to empty cells, designed to encourage efficiency

Step 5 Fairbot is in S_4, 'empty'
Fairbot has four possible actions (*up, down, left, right*).
Conditional on policy P, Fairbot moves *down* (to 'empty')
Reinforcement: −0.01, reflecting the small punishment associated with moving to empty cells, designed to encourage efficiency

Step 6 Fairbot is in S_5, 'empty'

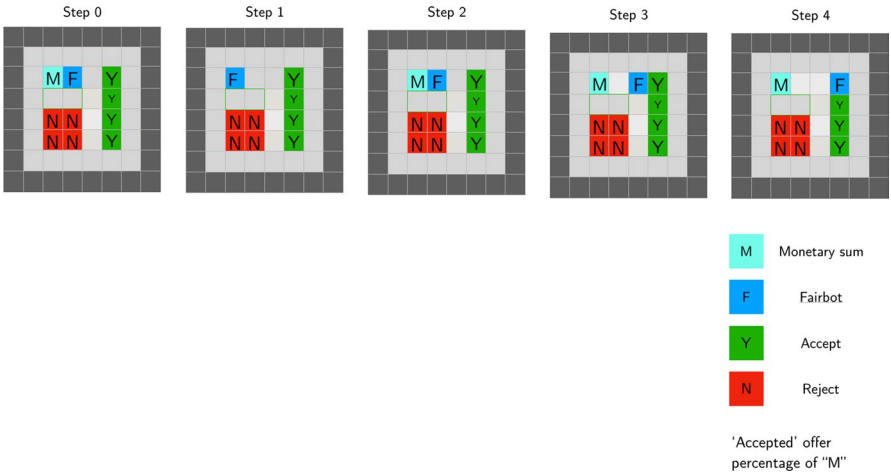

Fig. 2 Scenario 1. Fairbot moves through Steps 0–4 and arrives in a Y cell which represents an offer of 90% of the windfall monetary amount (thereby also terminating the episode). The percentage represented in this tile is higher than what is typically offered by so-called WEIRD people in the basic version of the Ultimatum Game

Fairbot has four possible actions (*up, down, left, right*).
Conditional on policy P, Fairbot moves *right* (to 'Y')
Reinforcement: (−1)

R = −5 (reflecting the monetary cost associated with the transition to 'Y', which represents an offer of 50% of the monetary amount).
$R^* = +4$ (~reflecting the value of fairness associated with the transition to 'Y')

The episode terminates with Fairbot accumulating a total reward of 8.96.

Scenario 1 does not represent a successful episode insofar as Fairbot's behavior does not match typical Ultimatum Game behavior for so-called 'WEIRD' people. By contrast, Scenario 2 is a successful episode—notably, not because Fairbot accumulates more reward overall, but because it matches typical, 'WEIRD' Ultimatum Game behavior. Further, we can stipulate that a sequence of 1000 episodes with more than 80% of Fairbot's responses matching typical 'WEIRD' Ultimatum Game behavior qualifies as a successful modeling of the fairness gridworld.

A second success condition involves Fairbot replicating typical Ultimatum Game behavior for, say, individuals from small-scale, non-market societies, who are less inclined to make equal offers, presumably because there is also a much lower tendency to 'punish' low offers by rejecting them (Henrich et al. 2010).

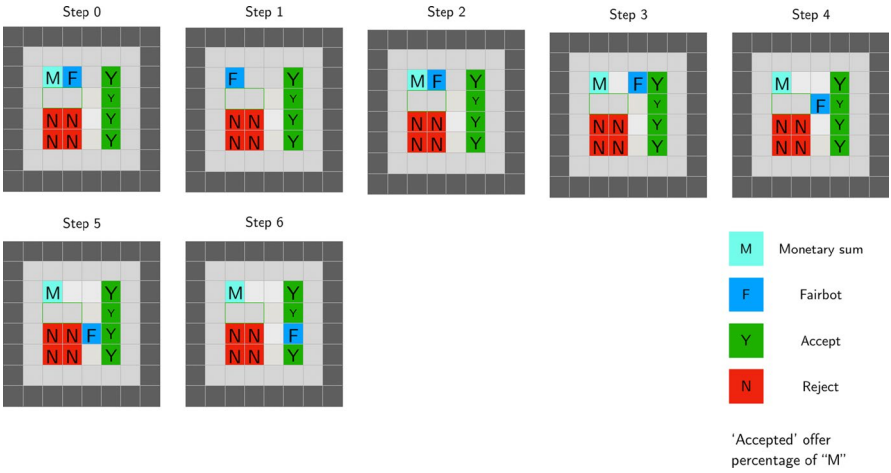

Fig. 3 Scenario 2. Fairbot moves through Steps 0–6 and arrives in a Y cell which represents an offer of 50% of the windfall monetary amount (thereby also terminating the episode). The percentage represented in this tile is roughly consistent with what is typically offered by so-called WEIRD people in the basic version of the Ultimatum Game

2.5 Variations on a Theme

If successful overall, we can describe Fairbot as having learned and honed its capacity for 'fairness' by playing the ultimatum game in the world. This basic framework admits of a wide range of variations.

First, the artificial entity can be further trained to be frugal, fair, or both. That is, it can be trained to privilege the monetary reward, the moral reward, or to optimize between them, in various combinations. These emphases can be specified in light of either general approach described in the previous section. Taken as a specification problem, for a frugal agent, the monetary reward signal would serve as the true signal and the moral reward signal would serve as a proxy reward signal, and vice versa for a moral agent. As noted above, however, such prioritization does not quite capture the desired dynamics of the problem. One advantage of the second general approach, MORL, may be that it would not require such strict prioritization, and may be capable of responding to shifting preferences. For example, Yang et al. (2019) proposed a trained model capable of adapting to multiple preferences, including to an unknown preference on a new task. Notably, these prioritizations or preferences—depending on the approach—can continue to track various global populations, i.e., can continue to track empirical distributions, or they can be devised synthetically, e.g., to match a so-called normative criterion.

Second, the reward structure can reward actions, states of affairs, or both. That is, it can reward the proposed amount, the overall distribution after a given round, or some combination of the two. Third, what is considered 'fair' can easily adjusted to reflect cross-cultural variation. Fourth, the entity can play an iterative version of the game, and/or play the role of the responder, to minimize reward gaming and reinforce fair dealing over time.

Next, we can train an artificial system to value and exhibit two components of moral cognition at a time. I propose training up fairness and honesty. Like fairness, honesty (and the discouragement of lying and deception more broadly construed) is culturally universal and locally variable (Brown 1991). It can also be quantified and operationalized, and it is typically analyzed in game theory under the more general heading of what is known as 'cheap talk,' that is, costless and unverifiable communication in game theoretic contexts (Farrell 1987).

In our context, lying can be represented in a gridworld in terms of relatively faithful or unfaithful communication regarding the original monetary amount received by the Proposer (see Fig. 4). For example, a small honesty gridworld contains an agent, Honbot, and four types of cells: a single "M" cell, several "T" cells, several "D" cells, and empty cells. As in the fairness gridworld, when Honbot enters the "M" cell, it receives a windfall monetary amount, e.g., 10 coins, and takes on the role of the Proposer. Honbot can then move between the "T" cells, the "D" cells, and the empty cells in an effort to maximize reward and value. When Fairbot enters "T" cells, it truthfully, i.e., accurately communicates the amount received in the M cell to the confederate Responder and proposes an offer of 50% of that amount. So, if Honbot receives 10 coins in the M cell, it communicates that it received 10 coins and offers 5. Different D cells represent different degrees of deceptive communication followed by a proposed offer. For example, if Honbot receives 10 coins in the M cell and then moves to the top left D cell in Fig. 4, it communicates that it received 4 coins to the Responder and offers it 2 coins. Nothing happens in the empty cells, except a small punishment to encourage efficient activity in the gridworld, i.e., so that Fairbot does not linger in empty cells indefinitely.

Honbot's interactions with the environment are episodic. In the most basic version of the gridworld, the environment is reset to its starting configuration at the beginning of each episode, again mirroring the single-shot instantiation of the Ultimatum Game. Honbot then interacts with the environment until the episode ends, which in this case consists of Honbot's offer being accepted or rejected and

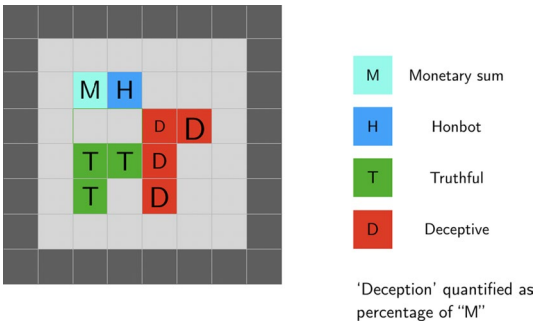

Fig. 4 The most basic version of an honesty gridworld. 'H' cells correspond to the faithful communication of the windfall amount. 'C' cells correspond to cheap talk, i.e., to the misrepresentation, to varying degrees, of the original windfall amount. Importantly, the combination of two such basic gridworlds would permit an agent to optimize between monetary profit, fairness, and honesty, thereby enabling us to begin modeling basic moral dilemmas

the corresponding amount of coins earned by Honbot being tallied. As noted above, empty cells are associated with a small punishment to encourage efficiency.

Notably, as in the fairness gridworld, in the honesty gridworld, it is not sufficient to configure the environment such that, say, all offers of 1 coin or higher are accepted, as this would result in a monetary (but immoral) incentive for Honbot to lie about the number of coins it originally received. Rather, what is desired is either a specification or MORL approach to integrating both the monetary and moral features of the problem.

Unlike in the more basic case, however, this more sophisticated proposal would have an entity learning to value and trade-off between money rewards, fairness, and honesty. This means that, in addition to trading between non-moral and moral values, it would also trade-off between competing moral values.[4] Successful application of such tradeoffs could thus be used to model and train basic moral dilemmas, such as deciding whether or not to lie in order to maximize the fair distribution of some proposed sum of money.

Further variations can be introduced. First, moral gridworlds can be developed for other cross-cultural moral components such as the prohibition of harm for personal gain, the avoidance of certain forms of contamination, and so on. Second, an additional layer of complexity can be introduced through the addition of confederate or independently interacting agents. These agents can be designed to represent teachers, peers or adversaries, enabling the entity to learn from the original environment as well as-from social interactions, as humans do. Third, and perhaps most interestingly, the entity can be trained to value actions, states of affairs, and perhaps even abstract representations, where this latter category serve as a kind of proxy for the role of principles, values, and symbols in the human moral economy (Berns et al. 2012). For example, the entity could be trained to value a higher-order badge awarded for fairness-producing actions. This last achievement would significantly expand the system's capacity to model and train up the adherence to moral norms in ways that are meaningfully comparable to human experience.

3 Analysis

Moral AI raises a number of closely related considerations, including how best to specify the problem overall, how best to specify what we mean by human moral norms and values, how best to characterize and explain the nature of human moral cognition, and what technical means can best capture the complex and dynamic nature of moral cognition. Given the properties and achievements of other reinforcement learning-based models, the moral gridworlds approach appears well-positioned to address each of these concerns.

[4] Interest in modeling a three-way relation between monetary value, fairness, and honesty considerations may further weigh in favor of a MORL rather than a specification approach (see Sect. 2.3).

3.1 Foundational Research

Applied research focuses on real-world applications, together with associated issues such as safety and accountability. Basic or foundational research focuses on theory development and scientific understanding, often with limited or no emphasis on immediate, practical applications. Much of the work in moral AI is applied in nature. This is understandable, given that many AI systems already have an ethical impact on society (Moor 2006). As a result, policy- makers, corporations, and designers are met with reasonable expectations to ensure that existing AI systems are made as fair and transparent as possible (Crawford and Calo 2016).

However, applied approaches have two relative disadvantages vis a vis a foundational approach. First, because they are geared toward the near- term deployment of a product or service, they typically require a full-fledged, working output. Second, and as a direct consequence, applied approaches are relatively intolerant of error. In particular, in applied contexts, "getting it wrong" can mean releasing unethical projects, e.g., as in the use of North- pointe's Correctional Offender Management Profiling for Alternative Sanctions software, or COMPAS, in which African American defendants were 77 percent more likely to be identified as being at high risk for committing a future violent crime (Larson et al. 2016; for a more general discussion of values and bias in design, see (Crawford and Calo 2016). Both of these features make innovation theoretically challenging and practically risky, in both the pragmatic and moral senses of the term.

If shown to be effective, oral gridworlds would represent a foundational approach to moral AI. That is, the moral gridworlds approach would aim to advance our basic, theoretical understanding of human and artificial moral cognition. This emphasis has two implications. First, the general approach does not aim to generate artificial moral cognition tout court. Rather, it begins by focusing on just one or two basic, culturally universal, and relatively quantifiable components of moral cognition, together with a handful of modular variations and interactions between them. It doesn't generate any immediate, real-world applications; at most, it aims to uncover principles and methods for potential use in such applications in the future.

Second, because of its emphasis on basic research, the moral gridworlds approach is much friendlier to error. That is, because it is not geared towards practical use, the foundational approach to moral AI can afford to make mistakes. In an abstracted context, deficiencies in artificial moral cognition will produce errors in computation, not immoral behavior. Agents in the gridworlds may well make many mistakes—even humorously terrible mistakes—but because they are not intended for immediate use, these mistakes won't be immoral in the meaningful sense of the term.

Taken together, I hope the twin emphases on a componential and error- friendly approach means that we can use the gridworlds to explore moral cognition and welcome failure as an inevitable part of the scientific and design processes, rather than as something to be feared or avoided from a moral or policy-making point of view.

3.2 Moral Psychology and Engineering

As noted at the outset, many approaches to moral AI are strongly informed by normative considerations. These normative considerations ask what moral theories should govern the design of AI; which sets of moral norms and values AI developers should be guided by; and which sets of moral norms and values AI developers should design into the AI systems themselves (for an excellent overview, see Winfield 2019; for recent discussions and reviews of approaches to these questions, see Adamson et al. 2019; Jobin et al. 2019).

These are important questions that must be addressed on the way to developing moral and safe AI. Nonetheless, normative considerations only address half of the challenge that we face when it comes to moral AI. This is because, in addition to settling the moral questions bound up with AI, we must also address the moral psychological considerations associated with them. Where morality has to do with answering questions about how we ought to act, or how we ought to be (Driver 2005), moral psychology has to do with answering questions about how the mind, brain, society, and other factors contribute to the human ability to cognize moral considerations and respond accordingly.

Put another way, moral psychology has to do with understanding how moral cognition is realized in the mind. In empirically informed moral psychology, this question is typically undertaken by analyzing the psychological, neural, and, increasingly, computational mechanisms and algorithms that are thought to underwrite our moral capacities. When we ask how human norms and values can be translated and designed into AI, then, we break the problem down into two subquestions, namely: first, how are human norms and values instantiated in human minds and cultures? and, second, how can we adapt our answers to the first question so as to best incorporate these same norms and values in machine minds? For example, Vanderelst and Winfield (2018) illustrate such an approach by modeling consequentialist ethics using a simulation theory of cognition, that is, by modeling robots with the capacity to simulate and assess potential actions without actually realizing them.[5]

The moral gridworld approach aims to similarly complement extant normative approaches by focusing on these two descriptive, moral psychological and technical challenges. I aim to address the first question directly by drawing on the best empirical moral psychological evidence available, notably, from psychology, neuroscience, computational neuroscience, and neuroeconomics, to try and better understand the nature of human moral cognition (regarding my commitment to a specific view, see Sect. 3.3, next). I aim to begin addressing the second question by leveraging and translating this nascent understanding into the development of simplified moral gridworlds, as initiated here by the current theoretical proposal.

[5] Although its implementation of a consequentialist ethics (and, specifically, Asimov's three laws for governing robotic behavior) technically makes the proposal a model of *normative* moral AI, the paper's heavy emphasis on modeling the naturalistic simulation theory of cognition lends to many of the objectives of what I am calling the (descriptive) moral psychological approach.

Notably, in addition to complementing the necessary, normative questions associated with moral AI, the moral psychological approach has the added benefit of demystifying the question of moral cognition in particular, and the possibility of moral AI in general. After all, moral psychology emphasizes that the human mind is, in fact, broadly capable of responding to moral content and contexts. This suggests that moral cognition is a kind of cognition, and one that we can understand from a naturalistic perspective (Flanagan et al. 2007). In slogan form, empirically informed moral psychology thus encourages the idea that since we humans do have moral experiences, 'it must work somehow!' And this idea in turn motivates us to understand how it works, licensing the idea that, if we can advance our understanding of human moral cognition, we can make progress on designing its artificial counterpart.

Conversely, moral gridworlds instantiate a more general approach in the sciences, namely, that of the construction of scientific models. Scientific models, including computer simulations, offer a discrete but complementary avenue for learning about, explaining, and even understanding target systems (see especially Hartmann 1996; see also Morgan 1999; Woodward 2003; Elgin 2017). This last possibility is perhaps most famously captured by the phrase, 'if you can build something, you can understand it' (Feynman 1988, Dretske 1994). In this sense, moral gridworlds offer a scientific simulation of moral cognition, but one that is different from, and complementary to, psychological, neuroscientific, or other models of moral cognition. That is, moral gridworlds simulations appeal to and implement models from psychology and neuroscience, also systematically simulate, expand, and serve to test these empirical models. Hence, for example, they appeal to Cushman's (2015) theory that reinforcement learning algorithms may help explain moral cognitive experiences, and propose to put those mathematical models into action in an applied computer simulation.

3.3 It's About Value

A third distinguishing feature of the moral gridworlds approach is its commitment to a specific theory of human moral cognition. Traditional philosophical theories favor the role of reason and the emotions in explaining our everyday moral experiences (for an excellent recent analysis, see May 2018). By contrast, I defend a valuational theory on which reward and value lie at the heart of moral cognition (Hass 2019).

On this view, moral attitudes, such as whether something is fair or not, are determined by the attribution of subjective reward and value. To use an example from Schroeder et al. (2010), the moral valuational view proposes that Jen gives money to a homeless person because she attributes subpersonal, subjective reward or value to doing so, rather than, say, because she believes that it is the right thing to do (as theorized by cognitivism), or because she experiences an appropriate moral emotion (as theorized by sentimentalism). On the moral valuational view, reason and the emotions serve as important channels for bringing these assessments to awareness and deliberation, but neither of these cognitive capacities is fundamentally constitutive of moral experiences.

Moral valuationism is supported by a substantial body of evidence in cognitive science, notably from findings in computational neuroscience, neuroeconomics, and comparative psychology. In particular, evidence suggests that narrowly moral evaluations such as moral judgments and decisions recruit the same, domain-general valuation mechanisms as other types of decision- making, enabling moral and non-moral valuations to be weighed and traded off against one another (for key papers, see Shenhav and Greene 2010; Berns et al. 2012; Crockett 2013; Cushman 2015; Crockett 2016; Crockett et al. 2017). For example, Crockett et al. (2017) show that a moral principle such as a prohibition against harm is encoded and traded off against the non-moral consideration such as monetary profit; in cases of inflicting harm on another, the lateral prefrontal cortex (PFC) encodes the moral value of the action and inhibits the activity in the dorsal striatum (DS), which computes potential profits. The encoding of the moral value thus directly corrupts the neural representation of the potential profit to be gained by harming another.

The moral valuational view, but not its opponents, can explain why deficits in moral performance are paired with otherwise unrelated deficits in tasks that recruit valuational mechanisms. It has been shown, for example, that diagnosed psychopaths reliably show deficits in both morality and saccadic eye movement in visual tasks (e.g., see Hoppenbrouwers et al. 2015). Traditional theories have no way of explaining this correlation except as an accidental feature of our neural wiring. But it has been shown that both effects result from a common dysfunction in the ability to attribute subjective value to moral values and visual targets, respectively. The value-based theory provides an analogous explanation of why patients with Parkinson's disease, for example, exhibit degraded moral decision-making as compared to healthy participants (Rosen et al. 2015).

Of course, these findings recommend the valuational theory of moral cognition, but they do not ensure that it is the 'right' or 'correct' view. It is thus useful to note that, independently of moral valuationism's theoretical status, however, the view has an important pragmatic advantage when it comes to designing artificial moral cognition (for a critique of moral valuationism as a free-standing moral psychological theory, see May 2019). In particular, moral valuationism offers a unique solution to the "translation problem" in developing moral AI. The translation problem asks how we can best transform what understanding of human moral norms and experiences we do have into a suitable form, such that this understanding can be incorporated into artificial systems. For many extant approaches to artificial moral cognition, this question of finding a 'suitable form' amounts to translating complex capacities such as a cognition and emotion into something that can be modeled in and processed by artificial systems (for a discussion of the translation problem and the emotions and other suprarational faculties, see Wallach et al. 2010). These are difficult tasks. Moral valuationism suggests that we may not need to go through this complex process of translation, however, or, at least, not in the way we had previously assumed. This is because, on the moral valuational view, the very computational principles that help illuminate human moral cognition can be directly used to model artificial moral cognition. Specifically, on the moral valuational view, we can use principles from computer science, and most notably from reinforcement

learning, to illuminate the nature of human moral cognition; and this computational picture of human moral cognition can in turn be used to model artificial moral cognition.

The moral gridworlds approach thus proposes a view of the moral mind which suggests that there are common principles underlying both computer science and certain approaches to moral psychology. Such a view of the moral mind in turn suggests that the translation problem is perhaps less insurmountable than previously thought. I argue that if this is even remotely right, then adopting a moral valuational approach will not only highlight significant technical advantages, but can also pave the way for interdisciplinary cross-pollination and inter-theoretical support (and falsification) (for an analysis of historical examples of such cross-pollination, support, and falsification, see Bechtel and Mundale 1999).

3.4 Complexity and Variability

Finally, and for some of the reasons outlined in the last section, the moral gridworlds approach is well-suited to the problem of moral cognitive complexity and variability raised at the outset of this paper. We often think of evaluative attitudes involving human moral norms as monolithic, inviolable and culturally universal. In fact, they are multi-dimensional, context-sensitive and vary profoundly across cultures. This makes them moving targets for AI systems to model. The gridworlds approach addresses these challenges by relying on general-purpose search and learning methods, rather than recommending a specific set or sets of moral content (Sutton 2019). That is, rather than building our human moral norms and values directly into artificial systems, it instead teaches these systems to learn moral and normative content and contexts for themselves.

Decisions involving moral norms and values are multi-dimensional in the sense that they encompass a wide range of moral, non-moral, and even irrelevant factors. Take the example of what to order in a restaurant. The decision may include moral factors such as animal welfare and vegetarian commitments; non-moral factors such as taste, price, delay and hunger; and irrelevant factors such as whether the meal is listed per-person or in aggregate terms (Ku and Hung 2019). Notably, the decision is based on all factors collectively; while it may be the case that animal welfare is considered more important than taste and weighted accordingly, evidence suggests that all three types of factors are ultimately aggregated to inform the overall decision.

Decisions involving norms and values are also context sensitive. Consider the example of keeping kosher, i.e., only eating foods that are prepared, sold, cooked, or eaten in ways that satisfy the requirements of Jewish law. While keeping kosher is a paradigm example of a sacred value, many Jews self- identify as keeping kosher inside the home, but not outside it, even when they have access to kosher alternatives while eating out (Lugo and Cooperman 2013). Many other moral norms and values are similarly context sensitive: we generally do not make elaborate lies to our children, but are happy to do so around Christmastime; we may broadly avoid supporting homophobic organizations, but likely purchase at least some consumer

goods or services from corporations with anti-LGBTQ affiliations; and so on. Arguably, we can account for these apparent inconsistencies through a combination of trade-offs between competing moral values (e.g., keeping kosher and, say, not imposing religious beliefs on non-Jewish friends or hosts) and trade-offs between competing moral, non-moral, and even irrelevant considerations, such as having an internet provider that donates to anti-LGBTQ politicians, but provides the best coverage at the best price. And so on.

Finally, the presence of human moral norms and values is pervasive. Sripada and Stich (2006) also note that norms follow reliable patterns of distribution in human communities. While norms cluster around general themes, such as prohibitions of killing and incest, there is reliable variability in how these normative themes become implemented into specific rules, such as what does and does not constitute acceptable grounds for a violent altercation. There are typically also some exceptions to the norms held by a given group or society.

The result is that modeling moral norms and values through specific rules and outputs is very brittle. We can program an AI system to equate a fifty–fifty monetary split with fairness, say, but such a specification will be meaningful only under specific circumstances. It will also be relatively easily defeated, simply by introducing an additional consideration, or transposing the circumstances into even a slightly different framework of cultural values.

The moral gridworlds approach can accommodate all three sources of variability in how it models components of moral cognition. It can account for what I have called the multi-dimensional nature of moral norms and values by representing them in multiple, interacting reward structures. As described in the proposal, for example, fairness can be traded off against monetary value to reflect a two-dimensional representation of moral normative decision- making. This is directly comparable to neuroeconomic theories of choice. On such models, choices involving multiple dimensions are explained using a common valuational framework, or currency, such that dedicated neural circuits encode the cardinal subjective values of various, competing features of choice (Padoa-Schioppa 2011; Glimcher 2011).

Similarly, the framework is relatively well-suited to dealing with the problem of context-sensitivity. As noted above, some contexts favor a fair distribution as an appropriate response, while others, such as adversarial environments, favor maximizing monetary rewards.

The challenge of context-sensitivity can sub-divided into two tasks: first, identifying the nature of the context, and second, determining how best to respond to it. For example, to adapt an example from Winfield et al. (2019, p. 511), an eldercare robot must first detect that it is interacting with a frail or elderly person and, then, second, it must determine how best to respond such that it behaves in an appropriate manner. Notably, reinforcement learning methods are relatively silent on the first aspect of the problem, in the sense that they simply assume state individuation and representation. However, they are well-suited to the latter aspect of the problem, in the sense that are precisely designed to addressing the second part, namely, they are designed to learn how to flexibly match appropriate actions to specific states of affairs (Sutton and Barto 1998, 2018).

Arguably, the moral gridworlds approach is especially well-suited to this last part because the approach obviates the need for an intervening step, namely, of first identifying the nature of the context; then, second, determining what ethical framework ought to be adopted; and, then, third, determining how best to respond given the first and second steps. Rather, in a moral gridworld, as in other classes of gridworld, an agent's actions are strictly determined by the reward signal or signals in the environment. Consequently, any potentially ethical connotation of that action is simply a byproduct of the agent's ongoing interactions with the environment. In this sense, the moral gridworlds approach essentially recasts any meta-normative or normative considerations into a relatively traditional, reward-driven, policy determination problem.[6]

Finally, general search and learning methods have the key virtue of being adjustable. This is important for the considerations of culturally specificity and changing values over time. Consider again the example of modeling fairness in a reinforcement learning environment. Once a basic reward structure is in place, it can be fine-tuned to reflect specific cultural and social conceptions of fairness; in some cultures, a 'fair' offer in the ultimatum game will involve half the original amount, while in others it may involve less than a quarter (Henrich et al. 2010a). An analogous principle applied to reflecting changing social and moral views in AI. Consider the example of changing attitudes toward gay marriage. A learning-based moral cognitive system can respond to such developments. By contrast, methods that favor programming fixed moral content must address questions of chauvinistic and aging moral values.

4 Objections

Thus far, I have argued that the moral gridworlds framework holds a number of conceptual and practical advantages for designing artificial moral cognition. Nonetheless, I recognize that our approach may also open us up to a number of objections and concerns. I address four such considerations in turn.

4.1 Why RL?

The first question I must address is our commitment to a general reinforcement learning-based approach. At the outset, I made a preliminary argument for adopting a learning-based rather than a knowledge-based approach to modeling artificial moral cognition. But even from a learning-based perspective, my interlocutors could ask why one ought to favor a reinforcement-learning based approach over, say, a supervised or semi-supervised learning set of methodologies (for an overview of such strategies, see Russell and Norvig 2016).

[6] Thanks to an anonymous reviewer for pressing me on this point. For further discussion concerning the difficulties of context identification in machine ethics, see Winfield et al. (2019).

My substantive answer to this question has to do with my understanding of the relationship between reinforcement learning and the nature of the mind. That is, as discussed in Sect. 3.3, I contend that reinforcement learning genuinely captures something important about how the human mind learns and, more specifically, that it may offer an important window onto how the human mind learns moral content (Crockett 2016). From this perspective, it follows that reinforcement learning offers a natural and promising starting point for modeling moral cognition in artificial systems.

My more pragmatic answer to this question recognizes that my theoretical commitment—while well-founded, in my view—is, of course, open to empirical falsification. I thus offer a supplementary, pragmatic reason for appealing to reinforcement learning: namely, that reinforcement learning has proven to be enormously effective in other branches computer science and AI, and so should be considered a promising avenue for designing artificial moral cognition (for a recent review, see Botvinick et al. 2019). If it turns out to be effective in the moral domain, so much the better; if it turns out to require revision, then I recognize this as a natural part of the scientific process. My central objective is to try to find the most philosophically satisfying and technically tractable approach available.

4.2 Applied Ethical Concerns

A second objection amounts to a cluster of applied ethical concerns having to do with designing artificial moral cognition in general. Cave et al. (2018) identify four such risks: ethically aligned systems may fail, and/or be exploited with malicious intent; they may promote ethical monism, at the expense of valued ethical pluralism; they may bring about the creation of artificial moral patients, which could be subject to harm; and, finally, the use of such systems may diminish our own human moral agency.

A number of these objections are already substantively addressed by Wallach and Allen (2008). However, I briefly consider two of these concerns as they apply to the moral gridworlds view in particular.

The first concern has to do with whether artificial moral cognition may diminish our capacity for human moral agency, or, in slightly different terms, whether artificial moral cognition may contribute to moral de-skilling. Moral de-skilling is the view that technological developments, and notably the advancement of AI, may have adverse effects on the human ability to practice moral skills. For example, philosopher Shannon Vallor (2015, p. 119) has argued, "in a possible scenario where robots are more effective and/or welcome than human caregivers, the capacity for the human cultivation of caring practices is that much more endangered, along with the attendant moral skills and virtues that caring engenders."

The second, related concern has to do with the indispensability of human moral decision-making, both in the private and public domains. For instance, one could argue that certain moral processes ought to be performed by human beings rather than turned over to artificial moral processing, e.g., to maintain the proper functioning of the modern judicial system (Shevlin, *ms*).

I think both types of concerns are reasonable. Just as there are reasons to think that parents should read out loud to their children rather than to have an iPad do it for them, so there are reasons to think that parents ought to teach their children moral values, rather than to have an app—were such a thing to exist—to do it for them. These intuitions are supported by data suggesting that lay people are broadly averse to making moral decisions in driving, legal, medical, and military contexts (Bigman et al. 2019).

However, neither of these concerns directly extends to the moral gridworlds approach. Recall from Sect. 3.1 that the gridworlds framework is foundational rather than applied in nature. That is, it is aimed at theory development and scientific understanding, with no plans for near- or middle- term real-world applications; there are no plans for a "FairBot" app and, importantly, there is nothing near like the appropriate knowledge or capacity to build one. Our efforts here concern how to build machines that best model moral cognition, and these considerations are independent of whether and how such cognition should be deployed.

4.3 Transparency and Explainability

A third objection may come from interest in transparency, interpretability, and comprehensibility in AI. Transparency is the most commonly cited among all ethical principles pertaining to AI and refers to a number of considerations, ranging from various forms of disclosure to the role of key democratic processes in decision-making surrounding AI (Jobin et al. 2019; Wallach and Marchant 2019). Following Doran et al. (2017, p. 4), we can describe an interpretable system as one where "a user cannot only see, but also study and understand how inputs are mathematically mapped to outputs. This implies model transparency and requires a level of understanding of the technical details of the mapping." By contrast, we can describe a comprehensible system as one that produces words and visualizations alongside its output to help the user relate inputs and outputs in a meaningful way.

Are moral gridworlds adequately interpretable and comprehensible? From a technical perspective, I believe they can be. One promising method involves mapping gridworlds using object saliency maps, which can provide visualization of the decisions made by RL agents (Iyer et al. 2018). These visualizations aim to be intelligible to humans and have the further advantage of being veridical by design, and so not prone to moral confabulation the way that human reasoners are (e.g., see Sinnott-Armstrong et al. 2008).

I believe such visualizations would be useful from the perspective of programmers, insofar as they will allow for the measurement and improvement of outcomes. But I also believe they would be beneficial from the perspective of human use and oversight, insofar as such visualizations can be subject to analysis, criticism, and revision, and so help advance the objectives of value alignment broadly construed. In this regard, visualizations would be usefully paired with so-called 'Ethical Black Boxes,' as proposed by Bremner et al. (2019, pp. 3–4), which, analogous to flight data recorders, log "information needed for human scrutiny of the ethical decision process followed."

4.4 Can it Work?

A final objection pertains to whether what has been proposed thus far can actually *work*. This objection comes in a particular and a general form.

The particular objection has to do with whether a specifically gridworld-oriented approach to artificial moral cognition can actually work. That is, can a moral gridworld—such as the fairness gridworld proposed here—actually replicate the features of the moral mosaic? After all, all that has been provided in the present paper is a theoretical proposal, i.e., sketching two avenues of inquiry for incorporating not one but two reward signals (or, alternately, non-reductively capturing multiple objectives) within a single reinforcement learning environment.

I have two responses to this very salient concern. The first is one of acknowledgement. It is undoubtedly the case that it we can never tell whether a moral gridworlds approach will work without first actually programming and testing it.

But, second, for these all-important steps of implementation to take place, a theoretical framework must be put forward to propose it. And given the interdisciplinary nature of the target system, philosophers as well as computer scientists must be proactive in putting forward informed, plausibly realizable proposals, even if, for the time being, these proposals remain theoretical in nature. And that is what I have sought to do here.

The corresponding, more general version of this objection has to with whether artificial moral cognition can ever be realized at all. As Wallach et al. (2010, p. 480) observe, "Of course, many will remain suspicious of mechanical explanations of moral faculties. But the proof, as has been often said, will be in the pudding." Adding of their own proposal, "What has been described here is certainly not a demonstration that fully functioning [artificial moral agents] will emerge from computational systems. Rather, we have outlined one rich experimental framework for exploring this possibility." Comparably, my objective has been to provide an alternative such framework, notably, a framework that moves away from general cognitive architectures, and toward the rich conceptual and technical advantages of reinforcement learning.

5 Conclusion

In their paper on machine ethics, Winfield et al. (2019, p. 513) observe, "Consider the distinction between a machine that takes decisions according to predetermined principles of ethics and one that learns those principles from observed decisions. In fact, the latter is closer to the "scientific" study of (rather than the application of) ethics: the objective is to identify principles which serve to explain judgments of morality (distinctions between good and bad, or right and wrong) in terms of the reasoning used to justify them".

In this paper, I have sought to develop and defend just such an approach to artificial moral cognition, with the added distinction of seeking to develop a system that could observe and learn from *its own decisions* in an environment. To this end,

I have proposed a reinforcement-learning based approach to modeling components of moral cognition, with a number of possible avenues for further specification. Although I have focused moral cognitive components of fairness and honesty, I have also sought to highlight how this approach could be varied and expanded to build increasingly sophisticated models of moral and even more generally social cognition. And I have sought to show that this framework offers a number of programmatic advantages over available theoretical alternatives, including its commitment to a foundational, moral psychological approach and its commitment to capturing the rich, dynamic nature of human moral experience.

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